Indian Institute of Information Technology Dharwad

Large Language Models

Mid Semester Examination, Date: September 19, 2026

Timing: 11.00 AM-12.30 PM | 7th Semester | Full Marks: 20

No marks will be awarded if all parts of a question are not answered together

1. Answer the following questions in brief: 4X2=8
   a. What is the key limitation of traditional n-gram language models?
   b. Why is positional encoding required in transformers?
   c. What is the difference between encoder-only and decoder-only architectures?
   d. What is masked attention and why is it used in GPT-like models?

2. Given the following:
   Query (Q) = [1, 0]
   Key (K1) = [1, 1], Key (K2) = [0, 1]
   Value (V1) = [2, 0], Value (V2) = [0, 3]
   a) Compute attention scores using dot product (Q·K)
   b) Apply softmax to obtain attention weights (approximate)
   c) Compute the final output vector. 2+2+2

3. Consider a language model predicting next-word probabilities:
   Word A: 0.4
   Word B: 0.3
   Word C: 0.2
   Word D: 0.1
   a) What word is selected using greedy decoding?
   b) If top-k (k=2) is used, which words are considered?
   c) Why is layer normalization applied twice in each transformer block? 1+1+1

4. For a training corpus: “I like NLP,” “I like AI,” “I love NLP,” use a bigram language model and Add-1 (Laplace) smoothing to:
   Using Add-1 smoothing, compute probability for: “I like ML.” 3
   Add <S> and </S> tokens.